

HCSM-09

Gauge Topology and Orientation Flip $\mathbb{Z}_2$: The Hodge Dual Sector Symmetry $\Omega^0 \leftrightarrow \Omega^2$ and Its Action on the Hypercharge Functional

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Abstract

The Hodge Complex Standard Model realizes the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet, and identifies the physical hypercharge functional as a specific element of the 15-dimensional sector-coarse space (HCSM-06, HCSM-07). This paper establishes the orientation-flip $\mathbb{Z}_2$ symmetry of this structure. We define the sector flip σ on the 56-dimensional Hodge sector $V_{56} = \Omega^0 \oplus \Omega^1 \oplus \Omega^2$ at $\lambda = 4$ by the swap $(\alpha_0, \omega_1, \alpha_2) \mapsto (\alpha_2, \omega_1, \alpha_0)$, induced by the Hodge star identification $\Omega^0 \cong \Omega^2$. We prove: (i) σ is an involution ($\sigma^2 = \mathbb{I}$); (ii) σ commutes with the D_4 action; (iii) under σ , the sector operators transform as $\mathbb{I} \rightarrow \mathbb{I}$, $\mathcal{F} \rightarrow \mathcal{F}$, $\mathcal{O} \rightarrow -\mathcal{O}$; (iv) the A_1 hypercharge coefficients transform as $(0, -1/6, -1/2) \rightarrow (0, -1/6, +1/2)$; (v) the B_1 coefficients transform as $(0, +1/2, +1/2) \rightarrow (0, +1/2, -1/2)$; (vi) the sector hypercharge values transform by swapping Ω^0 and Ω^2 ; (vii) the multiset of hypercharge magnitudes is invariant; (viii) the anomaly cancellation sums are invariant; (ix) if the CKM and PMNS matrices depend only on the multiset of hypercharge magnitudes, then their element magnitudes transform as $|V_{ij}| \rightarrow |V_{ji}|$ and $|U_{ij}| \rightarrow |U_{ji}|$ (multisets invariant) while the Jarlskog invariant flips sign. The flip exchanges the labels $u_R^c \leftrightarrow d_R^c$ on the A_1 block and $e_R^c \leftrightarrow \nu_R^c$ on the B_1 block, with no consequence for the magnitudes of any framework observable. The framework therefore does not privilege Ω^0 over Ω^2 : the assignment of one color-triplet multiplet to the outer sector and the other to the inner sector is a labeling convention. All claims are HCSM-native and verified at machine precision.

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1 Introduction

1.1 The orientation-flip question

The Hodge Complex Standard Model (HCSM) derives the gauge algebra

$$\mathfrak{g}_{\text{HCSM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5 \quad (1)$$

as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet (HCSM-06). The physical hypercharge functional is a specific element of the 15-dimensional sector-coarse space of the 56-dimensional Hodge sector (HCSM-07):

$$Y = \sum_{\rho} (\alpha_{\rho} \mathbb{I} + \beta_{\rho} \mathcal{F} + \gamma_{\rho} \mathcal{O}) P_{\rho}, \quad (2)$$

with A_1 coefficients $(0, -1/6, -1/2)$ and B_1 coefficients $(0, +1/2, +1/2)$, uniquely determined by the Standard Model content and the sector assignment.

The present paper asks: does the framework privilege the sector Ω^0 over the sector Ω^2 ? These two sectors carry identical D_4 decompositions $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$, and they are Hodge-dual to each other. The assignment of u_R^c to one and d_R^c to the other was fixed only by the ordering convention $Y(u_R^c) < Y(d_R^c)$.

We prove that the framework admits a $\mathbb{Z}_2$ symmetry, the orientation flip, that swaps $\Omega^0 \leftrightarrow \Omega^2$ and fixes Ω^1 , under which every physical prediction of the framework is invariant up to a relabeling of the color-triplet and lepton multiplets. The flip has no physical consequence for the magnitudes of any observable.

1.2 The one-line summary

The 56-dimensional Hodge sector admits a $\mathbb{Z}_2$ symmetry σ that swaps $\Omega^0 \leftrightarrow \Omega^2$ while fixing Ω^1 pointwise, induced by the Hodge star identification of the outer sectors. The sector operators transform as $\mathbb{I} \rightarrow \mathbb{I}$, $\mathcal{F} \rightarrow \mathcal{F}$, $\mathcal{O} \rightarrow -\mathcal{O}$. The hypercharge functional transforms by $u_R^c \leftrightarrow d_R^c$ on the A_1 block and $e_R^c \leftrightarrow \nu_R^c$ on the B_1 block. All magnitudes of physical predictions are invariant. The framework does not privilege Ω^0 over Ω^2 .

1.3 Main results

- (1) **The sector flip (Definition 3.1).** A $\mathbb{Z}_2$ action σ on the 56-dimensional Hodge sector $V_{56} = \Omega^0 \oplus \Omega^1 \oplus \Omega^2$ is defined by the swap $(\alpha_0, \omega_1, \alpha_2) \mapsto (\alpha_2, \omega_1, \alpha_0)$, induced by the Hodge star identification $\Omega^0 \cong \Omega^2$.
- (2) **Involution (Lemma 4.1).** $\sigma^2 = \mathbb{I}_{56}$.
- (3) **D_4 -equivariance (Theorem 5.1).** σ commutes with the D_4 action: $[\sigma, R] = [\sigma, S] = 0$.
- (4) **Sector operator transformation (Theorem 6.1).** Under σ ,

$$\sigma \mathbb{I} \sigma^{-1} = \mathbb{I}, \quad \sigma \mathcal{F} \sigma^{-1} = \mathcal{F}, \quad \sigma \mathcal{O} \sigma^{-1} = -\mathcal{O}. \quad (3)$$

- (5) **A_1 coefficient transformation (Theorem 7.1).** The A_1 hypercharge coefficients transform as

$$(0, -1/6, -1/2) \xrightarrow{\sigma} (0, -1/6, +1/2). \quad (4)$$

- (6) **B_1 coefficient transformation (Theorem 7.2).** The B_1 hypercharge coefficients transform as

$$(0, +1/2, +1/2) \xrightarrow{\sigma} (0, +1/2, -1/2). \quad (5)$$

- (7) **Sector value swap (Theorem 7.3).** The sector hypercharge values transform by swapping Ω^0 and Ω^2 :

$$Y_{A_1} : (-2/3, +1/6, +1/3) \xrightarrow{\sigma} (+1/3, +1/6, -2/3), \quad (6)$$

$$Y_{B_1} : (+1, -1/2, 0) \xrightarrow{\sigma} (0, -1/2, +1). \quad (7)$$

- (8) **Magnitude multiset invariance (Corollary 8.1).** The multiset of hypercharge magnitudes is invariant under σ .
- (9) **Anomaly preservation (Corollary 9.1).** The four one-loop anomaly cancellation conditions and the three cubic gauge anomaly cancellations are invariant under σ .
- (10) **Physical-prediction invariance (Corollary 8.3).** If a framework prediction depends only on the multiset of hypercharge magnitudes and on the D_4 structure of the multiplet, then it is invariant under σ up to the relabeling $u_R^c \leftrightarrow d_R^c$ and $e_R^c \leftrightarrow \nu_R^c$. In particular, the CKM magnitudes transform as $|V_{ij}| \rightarrow |V_{ji}|$ (multiset invariant), the PMNS magnitudes transform as $|U_{ij}| \rightarrow |U_{ji}|$ (multiset invariant), and the Jarlskog invariant J flips sign.
- (11) **Labeling convention (Corollary 9.3).** The sign of $\mathcal{O}$ is a labeling convention, fixed by the requirement $Y(u_R^c) < Y(d_R^c)$. Flipping the convention produces an equivalent formulation of the framework.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSM stack:

- HCSM-00: foundations, notation, and mapping.
- HCSM-01: the 8×8 torus substrate.
- HCSM-02: the Hodge complex; the Hodge star.
- HCSM-03: the Hodge star on $\ker D$; Lorentzian signature.
- HCSM-04: the 14-mode multiplet.
- HCSM-05: orbit decomposition.
- HCSM-06: sector-coarse correspondence.
- HCSM-07: hypercharge quantization.
- HCSM-08: anomaly cancellation.

1.5 Organization

Section 2 states the substrate, the Hodge complex, the Hodge star, the sector operators, and the hypercharge functional. Section 3 defines the sector flip σ . Section 4 proves it is an involution. Section 5 proves D_4 -equivariance. Section 6 derives the transformation of the sector operators. Section 7 derives the transformation of the hypercharge functional. Section 8 establishes the physical consequences. Section 9 establishes the anomaly preservation. Section 10 summarizes inputs and derived results. Section 11 gives reproducibility notes. Section 12 concludes.

2 Setup

2.1 The Hodge complex

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites (HCSM-01). The Hodge complex of differential forms is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (8)$$

of total dimension 256 (HCSM-02). The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree.

At the self-paired eigenvalue $\lambda = 4$, the multiplicities are 14 in Ω^0 , 28 in Ω^1 , and 14 in Ω^2 , giving the 56-dimensional Hodge sector

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (9)$$

with D_4 decomposition (HCSM-02, Theorem 7.1)

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (10)$$

Dimension check: $9 + 5 + 7 + 7 + 28 = 56$.

2.2 The Hodge star

The discrete Hodge star on the two-dimensional torus with standard orientation is

$$\star_0 : \Omega^0 \rightarrow \Omega^2, \quad \star_0 = \text{id}; \quad \star_1 : \Omega^1 \rightarrow \Omega^1, \quad (\star_1 \omega)_x = \omega_y, \quad (\star_1 \omega)_y = -\omega_x; \quad \star_2 : \Omega^2 \rightarrow \Omega^0, \quad \star_2 = \text{id}. \quad (11)$$

In block form on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$,

$$\star = \begin{pmatrix} 0 & 0 & \star_0 \\ 0 & \star_1 & 0 \\ \star_2 & 0 & 0 \end{pmatrix}. \quad (12)$$

The Hodge star satisfies $\star_0^2 = +\mathbb{I}_{\Omega^0}$, $\star_1^2 = -\mathbb{I}_{\Omega^1}$, $\star_2^2 = +\mathbb{I}_{\Omega^2}$, and commutes with $D^2 = \Delta$ (HCSM-03).

2.3 The D_4 action

The dihedral group D_4 of order 8 acts on Λ by the point group of the square, generated by the rotation $r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8}$ and the reflection $s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}$, subject to $r^4 = s^2 = (rs)^2 = e$.

2.4 Sector operators

Definition 2.1 (Sector operators). *On V_{56} , define the three sector operators*

$$\mathbb{I} = P_{\Omega^0} + P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{F} = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{O} = P_{\Omega^0} - P_{\Omega^2}. \quad (13)$$

Restricted to the three sectors $(\Omega^0, \Omega^1, \Omega^2)$, their signatures are

$$\mathbb{I}|_{\text{sectors}} = (1, 1, 1), \quad \mathcal{F}|_{\text{sectors}} = (1, -1, 1), \quad \mathcal{O}|_{\text{sectors}} = (1, 0, -1). \quad (14)$$

2.5 The hypercharge functional

The hypercharge functional is a specific element of the 15-dimensional sector-coarse space (HCSM-07):

$$Y = \sum_{\rho} (\alpha_{\rho} \mathbb{I} + \beta_{\rho} \mathcal{F} + \gamma_{\rho} \mathcal{O}) P_{\rho}, \quad (15)$$

with A_1 and B_1 coefficients uniquely determined by the Standard Model content and the sector assignment:

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -1/6, -1/2), \quad (\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +1/2, +1/2). \quad (16)$$

The corresponding sector hypercharge values are

$$Y_{A_1} : (-2/3, +1/6, +1/3), \quad Y_{B_1} : (+1, -1/2, 0). \quad (17)$$

3 The sector flip σ

Definition 3.1 (Sector flip). *The sector flip $\sigma : V_{56} \rightarrow V_{56}$ is the $\mathbb{R}$ -linear map*

$$\sigma(\alpha_0 + \omega_1 + \alpha_2) = \star_0 \alpha_2 + \omega_1 + \star_2 \alpha_0, \quad (18)$$

where $\alpha_0 \in \Omega^0|_{\lambda=4}$, $\omega_1 \in \Omega^1|_{\lambda=4}$, $\alpha_2 \in \Omega^2|_{\lambda=4}$, and $\star_0, \star_2$ are the Hodge star maps.

Remark 3.2 (Simplification). *Because $\star_0$ and $\star_2$ are the identity maps on their respective sectors (HCSM-02), the flip takes the simple form*

$$\sigma(\alpha_0, \omega_1, \alpha_2) = (\alpha_2, \omega_1, \alpha_0). \quad (19)$$

That is, σ swaps the Ω^0 and Ω^2 sectors while fixing Ω^1 pointwise.

Remark 3.3 (Topological character). *The flip σ is defined using the Hodge star, which depends only on the orientation of the torus, not on a metric. It is therefore a topological symmetry: it is insensitive to continuous deformations of the metric and depends only on the cohomological (de Rham) structure of the Hodge complex. This is the “gauge topology” content of the paper’s title.*

4 Involution

Lemma 4.1 (Involution). $\sigma^2 = \mathbb{I}_{56}$.

Proof. For $v = (\alpha_0, \omega_1, \alpha_2) \in V_{56}$,

$$\sigma^2(v) = \sigma(\alpha_2, \omega_1, \alpha_0) = (\alpha_0, \omega_1, \alpha_2) = v. \quad (20)$$

Hence σ is an involution. $\square$

Corollary 4.2 ($\mathbb{Z}_2$ action). σ generates a $\mathbb{Z}_2$ action on V_{56} : the group $\mathbb{Z}_2 = \{e, \sigma\}$ acts on V_{56} by $e \cdot v = v$ and $\sigma \cdot v = \sigma(v)$.

Remark 4.3 (Non-degeneracy). $\sigma \neq \mathbb{I}_{56}$, since σ exchanges nontrivial subspaces $\Omega^0|_{\lambda=4} \neq \Omega^2|_{\lambda=4}$ (as subspaces of V_{56}). The $\mathbb{Z}_2$ action is therefore faithful.

5 D_4 -equivariance

Theorem 5.1 (D_4 -equivariance). *The sector flip commutes with the D_4 action on V_{56} :*

$$[\sigma, R] = [\sigma, S] = 0. \quad (21)$$

Proof. It suffices to check the two generators $R = \rho(r)$ and $S = \rho(s)$ on each sector.

On $\Omega^0 \oplus \Omega^2$. The D_4 action on Ω^0 and Ω^2 are site-permutation actions induced by the point group of the square. Because the Hodge star identification $\Omega^0 \cong \Omega^2$ is D_4 -equivariant (HCSM-02, Proposition 2.5), the two actions coincide under this identification. Hence for $v = (\alpha_0, 0, \alpha_2)$,

$$\sigma Rv = \sigma(R\alpha_0, 0, R\alpha_2) = (R\alpha_2, 0, R\alpha_0) = R\sigma v. \quad (22)$$

The same computation holds for S .

On Ω^1 . σ acts trivially on Ω^1 , so

$$\sigma Rv = \sigma(R\omega_1) = R\omega_1 = R\sigma v \quad (23)$$

for $v = (0, \omega_1, 0)$. The same holds for S .

Combining the two cases, σ commutes with R and S on V_{56} . $\square$

Corollary 5.2 (Sector-coarse compatibility). *The flip σ preserves the D_4 isotypic structure of V_{56} . In particular, σ commutes with the isotypic projectors P_ρ for all $\rho \in \{A_1, A_2, B_1, B_2, E\}$.*

6 Sector-operator transformation

Theorem 6.1 (Sector-operator transformation). *Under the flip σ , the sector operators transform as*

$$\sigma \mathbb{I} \sigma^{-1} = \mathbb{I}, \quad \sigma \mathcal{F} \sigma^{-1} = \mathcal{F}, \quad \sigma \mathcal{O} \sigma^{-1} = -\mathcal{O}. \quad (24)$$

Proof. The sector projections transform under σ as

$$\sigma P_{\Omega^0} \sigma^{-1} = P_{\Omega^2}, \quad \sigma P_{\Omega^1} \sigma^{-1} = P_{\Omega^1}, \quad \sigma P_{\Omega^2} \sigma^{-1} = P_{\Omega^0}. \quad (25)$$

(Conjugating P_{Ω^0} by σ gives the projector onto the image $\sigma(\Omega^0) = \Omega^2$, and similarly for the other sectors.)

Then:

$$\sigma \mathbb{I} \sigma^{-1} = \sigma P_{\Omega^0} \sigma^{-1} + \sigma P_{\Omega^1} \sigma^{-1} + \sigma P_{\Omega^2} \sigma^{-1} = P_{\Omega^2} + P_{\Omega^1} + P_{\Omega^0} = \mathbb{I}, \quad (26)$$

$$\sigma \mathcal{F} \sigma^{-1} = \sigma P_{\Omega^0} \sigma^{-1} - \sigma P_{\Omega^1} \sigma^{-1} + \sigma P_{\Omega^2} \sigma^{-1} = P_{\Omega^2} - P_{\Omega^1} + P_{\Omega^0} = \mathcal{F}, \quad (27)$$

$$\sigma \mathcal{O} \sigma^{-1} = \sigma P_{\Omega^0} \sigma^{-1} - \sigma P_{\Omega^2} \sigma^{-1} = P_{\Omega^2} - P_{\Omega^0} = -\mathcal{O}. \quad (28)$$

□

Corollary 6.2 (Algebraic transformation). *The flip σ acts on the sector-coarse algebra as the involution*

$$(\alpha, \beta, \gamma) \xrightarrow{\sigma} (\alpha, \beta, -\gamma) \quad (29)$$

on the coefficients of $\alpha \mathbb{I} + \beta \mathcal{F} + \gamma \mathcal{O}$.

Proof. Immediate from Theorem 6.1: the span of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ is preserved by σ , with the stated action on coordinates. □

7 Hypercharge functional transformation

Theorem 7.1 (A_1 coefficient transformation). *Under σ , the A_1 hypercharge coefficients transform as*

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -1/6, -1/2) \xrightarrow{\sigma} (0, -1/6, +1/2). \quad (30)$$

Proof. The flip acts on the sector-coarse algebra as $(\alpha, \beta, \gamma) \mapsto (\alpha, \beta, -\gamma)$. Applying this to the A_1 coefficients:

$$(0, -1/6, -1/2) \mapsto (0, -1/6, +1/2). \quad (31)$$

The projectors P_ρ are invariant under σ (Theorem 5.1), so the action on the full functional is well-defined. □

Theorem 7.2 (B_1 coefficient transformation). *Under σ , the B_1 hypercharge coefficients transform as*

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +1/2, +1/2) \xrightarrow{\sigma} (0, +1/2, -1/2). \quad (32)$$

Proof. Same as Theorem 7.1 applied to the B_1 coefficients. □

Theorem 7.3 (Sector-value swap). *Under σ , the sector hypercharge values transform by swapping Ω^0 and Ω^2 :*

$$Y_{A_1} : (-2/3, +1/6, +1/3) \xrightarrow{\sigma} (+1/3, +1/6, -2/3), \quad (33)$$

$$Y_{B_1} : (+1, -1/2, 0) \xrightarrow{\sigma} (0, -1/2, +1). \quad (34)$$

Proof. Direct evaluation of the transformed coefficients on the sector signatures.

For A_1 with transformed coefficients $(0, -1/6, +1/2)$:

$$Y'_{A_1}(\Omega^0) = 0 + (-1/6) + (+1/2) = +1/3, \quad (35)$$

$$Y'_{A_1}(\Omega^1) = 0 - (-1/6) = +1/6, \quad (36)$$

$$Y'_{A_1}(\Omega^2) = 0 + (-1/6) - (+1/2) = -2/3. \quad (37)$$

So Y_{A_1} on $(\Omega^0, \Omega^1, \Omega^2)$ transforms from $(-2/3, +1/6, +1/3)$ to $(+1/3, +1/6, -2/3)$.

For B_1 with transformed coefficients $(0, +1/2, -1/2)$:

$$Y'_{B_1}(\Omega^0) = 0 + (+1/2) + (-1/2) = 0, \quad (38)$$

$$Y'_{B_1}(\Omega^1) = 0 - (+1/2) = -1/2, \quad (39)$$

$$Y'_{B_1}(\Omega^2) = 0 + (+1/2) - (-1/2) = +1. \quad (40)$$

So Y_{B_1} transforms from $(+1, -1/2, 0)$ to $(0, -1/2, +1)$.

In both cases, the transformation is exactly the swap $\Omega^0 \leftrightarrow \Omega^2$. $\square$

8 Physical consequences

8.1 Magnitude-multiset invariance

Corollary 8.1 (Magnitude multiset invariant). *The multiset of hypercharge magnitudes*

$$\{|Y_{A_1}(\Omega^0)|, |Y_{A_1}(\Omega^1)|, |Y_{A_1}(\Omega^2)|\} = \{2/3, 1/6, 1/3\} \quad (41)$$

on the A_1 block and

$$\{|Y_{B_1}(\Omega^0)|, |Y_{B_1}(\Omega^1)|, |Y_{B_1}(\Omega^2)|\} = \{1, 1/2, 0\} \quad (42)$$

on the B_1 block are invariant under σ .

Proof. By Theorem 7.3, the flip swaps Ω^0 and Ω^2 while fixing Ω^1 . On the A_1 block, the sector values $\{-2/3, +1/6, +1/3\}$ map to $\{+1/3, +1/6, -2/3\}$, which has the same multiset of absolute values. On the B_1 block, $\{+1, -1/2, 0\}$ maps to $\{0, -1/2, +1\}$, again with the same multiset of absolute values. $\square$

8.2 Relabeling of the color-triplet and lepton multiplets

Corollary 8.2 (Multiplet relabeling). *Under σ , the color-triplet multiplet assignment transforms as*

$$u_R^c \leftrightarrow d_R^c, \quad (43)$$

and the lepton-multiplet assignment transforms as

$$e_R^c \leftrightarrow \nu_R^c. \quad (44)$$

The multiplet Q_L (color-triplet, weak doublet) and L_L (color-singlet, weak doublet), both on Ω^1 , are invariant.

Proof. By Theorem 7.3, σ swaps Ω^0 and Ω^2 . Under the sector assignment (Input **I5.**), u_R^c occupies Ω^0 and d_R^c occupies Ω^2 ; hence they are swapped. Similarly for e_R^c and ν_R^c on the B_1 block (which carry the sector values $+1$ on Ω^0 and 0 on Ω^2). $\square$

8.3 Invariance of physical-prediction magnitudes

Corollary 8.3 (Physical-prediction invariance). *Any framework prediction that depends only on (i) the multiset of hypercharge magnitudes and (ii) the D_4 structure of the multiplet is invariant under σ up to the relabeling $u_R^c \leftrightarrow d_R^c$ and $e_R^c \leftrightarrow \nu_R^c$. In particular:*

(a) CKM magnitudes. *The magnitudes of the CKM matrix transform as*

$$|V_{ij}| \xrightarrow{\sigma} |V_{ji}|. \quad (45)$$

The multiset $\{|V_{ij}|\}$ is invariant. The equality $|V_{ij}| = |V_{ji}|$ does not hold in general; the transformation is a relabeling of the quark generations, not an identity on individual elements.

(b) PMNS magnitudes. *The magnitudes of the PMNS matrix transform as*

$$|U_{ij}| \xrightarrow{\sigma} |U_{ji}|. \quad (46)$$

The multiset $\{|U_{ij}|\}$ is invariant.

(c) Jarlskog invariant. *The Jarlskog invariant J flips sign:*

$$J = c_{12}s_{12}c_{23}s_{23}c_{13}^2s_{13}\sin\delta \xrightarrow{\sigma} -J. \quad (47)$$

The magnitude $|J|$ is preserved.

Proof. Under $u \leftrightarrow d$ exchange, the CKM matrix transforms as $V \rightarrow V^\dagger$. The magnitudes satisfy $|(V^\dagger)_{ij}| = |V_{ji}|$. The multiset of magnitudes is preserved because the map $(i, j) \mapsto (j, i)$ is a bijection on the index set. The Jarlskog invariant, in the standard PDG parametrization, is $J = c_{12}s_{12}c_{23}s_{23}c_{13}^2s_{13}\sin\delta$. Under complex conjugation of the CKM matrix (equivalent to the $u \leftrightarrow d$ exchange combined with a phase redefinition of the quark fields), $\delta \rightarrow -\delta$, so $\sin\delta \rightarrow -\sin\delta$. Hence $J \rightarrow -J$. The magnitude $|J|$ is preserved.

The same argument applies to the PMNS matrix U under $e \leftrightarrow \nu$ exchange. $\square$

Remark 8.4 (Status of the flavor-sector application). *The flavor-sector predictions (CKM and PMNS matrices) are the subject of HCSM-34 and HCSM-36. The corollary above is stated as a symmetry of the HCSM sector-coarse structure, and applies to any framework prediction that inherits its dependence on the SM content through the hypercharge multiset.*

The corollary does not claim that CKM and PMNS are derived in HCSM-09; it claims that if they are derived in the flavor sector with the standard dependence on hypercharge magnitudes, then σ acts on them as $V \rightarrow V^\dagger$ and $U \rightarrow U^\dagger$.

9 Anomaly preservation

Corollary 9.1 (Anomaly preservation). *The four one-loop anomaly cancellation conditions*

$$\text{Tr}(Y) = 0, \quad \text{Tr}(Y^3) = 0, \quad \text{SU}(2)^2\text{U}(1) = 0, \quad \text{SU}(3)^2\text{U}(1) = 0 \quad (48)$$

and the three cubic gauge anomaly cancellations are invariant under σ .

Proof. Under σ , the sector hypercharge values are permuted: Ω^0 and Ω^2 exchange their values, and Ω^1 is fixed. The anomaly sums are weighted traces over the sector content; the weights depend on the SM representation multiplicities attached to each sector.

For the mixed $\text{SU}(3)^2\text{U}(1)$ anomaly, the sector-coarse form is (HCSM-08, Proposition 18)

$$Y_0 + 2Y_1 + Y_2 = 0. \quad (49)$$

Under σ , this becomes

$$Y_2 + 2Y_1 + Y_0 = 0. \quad (50)$$

Since the sum is symmetric in Y_0 and Y_2 , the condition is preserved.

For the other three conditions, the invariance follows from the multiset invariance of Corollary 8.1: any symmetric function of the sector hypercharge values that vanishes before σ also vanishes after. Specifically, $\text{Tr}(Y)$, $\text{Tr}(Y^3)$, and $\text{SU}(2)^2\text{U}(1)$ are all symmetric sums of hypercharge values weighted by sector content, and the flip permutes Ω^0 and Ω^2 without changing the multiset of values.

The cubic gauge anomalies vanish by content structure (HCSM-08, Theorem 11) regardless of the sector assignment, so they are trivially invariant. $\square$

Remark 9.2 (Physical reading). *The anomaly preservation means that if the HCSM hypercharge functional is anomaly-free with the assignment $u_R^c \in \Omega^0$, $d_R^c \in \Omega^2$, it is also anomaly-free with the flipped assignment $u_R^c \in \Omega^2$, $d_R^c \in \Omega^0$. The anomaly cancellation does not distinguish the two assignments.*

Corollary 9.3 (Labeling convention). *The sign of $\mathcal{O}$ is a labeling convention, fixed by the requirement $Y(u_R^c) < Y(d_R^c)$. Flipping the convention produces an equivalent formulation of the framework.*

Proof. Immediate from Corollary 8.3 and Remark 9.2. The two assignments produce identical physical predictions up to the relabeling $u_R^c \leftrightarrow d_R^c$, $e_R^c \leftrightarrow \nu_R^c$. $\square$

10 Inputs and derived

10.1 Derived

The following results are proved in this paper:

- (1) The sector flip σ is a well-defined $\mathbb{Z}_2$ action on V_{56} (Definition 3.1, Lemma 4.1).
- (2) σ commutes with D_4 (Theorem 5.1).
- (3) σ acts on the sector operators as $\mathbb{I} \rightarrow \mathbb{I}$, $\mathcal{F} \rightarrow \mathcal{F}$, $\mathcal{O} \rightarrow -\mathcal{O}$ (Theorem 6.1).
- (4) The A_1 coefficients transform as $(0, -1/6, -1/2) \rightarrow (0, -1/6, +1/2)$ (Theorem 7.1).
- (5) The B_1 coefficients transform as $(0, +1/2, +1/2) \rightarrow (0, +1/2, -1/2)$ (Theorem 7.2).
- (6) The sector hypercharge values swap $\Omega^0 \leftrightarrow \Omega^2$ (Theorem 7.3).
- (7) The magnitude multiset is invariant (Corollary 8.1).
- (8) The relabeling is $u_R^c \leftrightarrow d_R^c$, $e_R^c \leftrightarrow \nu_R^c$ (Corollary 8.2).
- (9) Anomaly cancellation is preserved (Corollary 9.1).
- (10) Physical-prediction magnitudes are invariant; the Jarlskog invariant flips sign (Corollary 8.3).
- (11) The orientation of $\mathcal{O}$ is a labeling convention (Corollary 9.3).

10.2 Inputs

- I1.** HCSM-02, HCSM-03: the Hodge complex, the Hodge star, and the D_4 -equivariance of the Hodge star identification $\Omega^0 \cong \Omega^2$.
- I2.** HCSM-06: the sector-coarse space and the sector operators.
- I3.** HCSM-07: the hypercharge coefficients and the sector hypercharge values.

- I4.** HCSM-08: the anomaly cancellation conditions.
- I5. Sector assignment convention:** the assignment $u_R^c \in \Omega^0$, $Q_L \in \Omega^1$, $d_R^c \in \Omega^2$ from HCSM-07, with the ordering convention $Y(u_R^c) < Y(d_R^c)$. This is a labeling convention (Corollary 9.3), not a physical input.
- I6.** Universal tools: the transformation of the CKM matrix $V \rightarrow V^\dagger$ under $u \leftrightarrow d$ exchange, the transformation of the PMNS matrix $U \rightarrow U^\dagger$ under $e \leftrightarrow \nu$ exchange, and the standard PDG parametrization of the Jarlskog invariant.

10.3 What is not claimed

We do not claim to derive the CKM or PMNS matrices. These are the subject of HCSM-34 and HCSM-36. We claim that if the flavor-sector predictions inherit their dependence on the SM content through the multiset of hypercharge magnitudes, then they are invariant under σ up to the relabeling.

We do not claim to privilege any specific sector assignment. The framework is invariant under σ : the assignment $u_R^c \in \Omega^0$, $d_R^c \in \Omega^2$ and the flipped assignment $u_R^c \in \Omega^2$, $d_R^c \in \Omega^0$ produce equivalent formulations. The convention $Y(u_R^c) < Y(d_R^c)$ is a labeling convention, not a physical content.

11 Reproducibility

All numerical and algebraic claims in this paper are verified at machine precision in IEEE-754 double precision, or exactly in rational arithmetic. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-09_verification.py`. It constructs the Hodge complex on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, the sector operators, and the hypercharge functional, and verifies:

- (1) The sector flip σ as an explicit $\mathbb{Z}_2$ action on V_{56} (Lemma 4.1).
- (2) D_4 -equivariance $[\sigma, R] = [\sigma, S] = 0$ (Theorem 5.1).
- (3) The sector-operator transformation (Theorem 6.1).
- (4) The coefficient transformations of Theorems 7.1 and 7.2.
- (5) The sector-value swap of Theorem 7.3.
- (6) The magnitude-multiset invariance (Corollary 8.1).
- (7) The anomaly preservation (Corollary 9.1).
- (8) The physical-prediction invariance of Corollary 8.3.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

12 Conclusion

This paper has established the orientation-flip $\mathbb{Z}_2$ symmetry of the HCSM sector-coarse structure. The flip σ , defined by the Hodge-star identification of the outer sectors $\Omega^0 \cong \Omega^2$, is a well-defined involution on the 56-dimensional Hodge sector that commutes with D_4 and swaps $\Omega^0 \leftrightarrow \Omega^2$ while fixing Ω^1 .

Under σ , the sector operators transform as $\mathbb{I} \rightarrow \mathbb{I}$, $\mathcal{F} \rightarrow \mathcal{F}$, and $\mathcal{O} \rightarrow -\mathcal{O}$. The hypercharge coefficients transform as $(0, -1/6, -1/2) \rightarrow (0, -1/6, +1/2)$ on the A_1 block and $(0, +1/2, +1/2) \rightarrow (0, +1/2, -1/2)$ on the B_1 block. The sector hypercharge values swap $\Omega^0 \leftrightarrow \Omega^2$, exchanging $u_R^c \leftrightarrow d_R^c$ and $e_R^c \leftrightarrow \nu_R^c$.

All framework predictions of the magnitudes of physical observables are invariant under σ . The hypercharge magnitude multiset is preserved, the anomaly cancellation conditions are preserved, and (conditional on the flavor-sector dependence) the CKM and PMNS element magnitudes transform as $|V_{ij}| \rightarrow |V_{ji}|$ and $|U_{ij}| \rightarrow |U_{ji}|$ (multisets invariant) while the Jarlskog invariant flips sign. The framework does not privilege Ω^0 over Ω^2 : the two outer sectors are Hodge-dual, carry identical D_4 decompositions, and are distinguished only by the labeling convention for the orientation operator $\mathcal{O}$.

The framework derives the space of hypercharge functionals (HCSM-06), the coefficient values (HCSM-07), the anomaly cancellation (HCSM-08), and the $\mathbb{Z}_2$ symmetry that acts on all of them (this paper). The $\mathbb{Z}_2$ symmetry closes the gauge-sector structure of the framework.

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A Web Crawler Metadata

A.1 Dublin Core

- `DC.title`: HCSM-09: Gauge Topology and Orientation-Flip $\mathbb{Z}_2$
- `DC.creator`: Preschutti, Stanley
- `DC.creator.orcid`: 0009-0004-5445-1744
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- `DC.rights`: All rights reserved
- `DC.subject`: Hodge Complex Standard Model; HCSM; orientation-flip $\mathbb{Z}_2$; sector flip; Hodge star; $\Omega^0 \leftrightarrow \Omega^2$ swap; hypercharge functional; CKM Jarlskog; anomaly preservation; labeling convention
- `DC.description`: The Hodge Complex Standard Model admits a $\mathbb{Z}_2$ symmetry, the orientation flip, defined by the Hodge-star identification of the outer sectors Ω^0 and Ω^2 . The flip swaps $\Omega^0 \leftrightarrow \Omega^2$ while fixing Ω^1 , commutes with D_4 , and transforms the sector operators as $\mathbb{I} \rightarrow \mathbb{I}$, $\mathcal{F} \rightarrow \mathcal{F}$, $\mathcal{O} \rightarrow -\mathcal{O}$. The hypercharge coefficients transform as $(0, -1/6, -1/2) \rightarrow (0, -1/6, +1/2)$ on A_1 and $(0, +1/2, +1/2) \rightarrow (0, +1/2, -1/2)$ on B_1 . The sector values swap $\Omega^0 \leftrightarrow \Omega^2$. The multiset of hypercharge magnitudes is invariant; anomaly cancellation is preserved; CKM/PMNS magnitudes transform as $|V_{ij}| \rightarrow |V_{ji}|$ with invariant multisets; the Jarlskog invariant flips sign. The framework does not privilege Ω^0 over Ω^2 .
- `DC.identifier`: HCSM-09-2026-09-Rev1

A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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    anomaly preservation, labeling convention},
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    identification of the outer sectors  $\Omega^0$  and  $\Omega^2$ . The flip
    swaps  $\Omega^0 \leftrightarrow \Omega^2$  while fixing  $\Omega^1$ , commutes with  $D_4$ ,
    and transforms the sector operators as  $I \rightarrow I$ ,  $F \rightarrow F$ ,  $0 \rightarrow -0$ .
    The hypercharge coefficients transform as  $(0, -1/6, -1/2) \rightarrow$ 
 $(0, -1/6, +1/2)$  on  $A_1$  and  $(0, +1/2, +1/2) \rightarrow (0, +1/2, -1/2)$  on  $B_1$ .
    The framework does not privilege  $\Omega^0$  over  $\Omega^2$ .}
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